

The Yoneda Lemma

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1 Introduction

This write up aims to summarize what I learned during the Fall 2024 DRP under mentorship of Gautam Manohar. The primary focus was exploring Category Theory, where we ultimately converged into learning about the Yoneda Lemma and some of its applications. I largely relied on a wide array of sources, include Math3ma's blog posts, NLab, various YouTube Channels, and the weekly sessions with my mentor. This write up does assume some of the background I came in with (such as basic algebra, group theory, etc). However, additional terms will be defined where I see fit.

Category Theory is a study of metamathematics, a powerful tool that uncovers rich connections between different areas of mathematics. It can be thought of as an extension of Set theory, but at the same time is distinguished by the tools it provides.

1.1 Overarching concepts

There are two concepts in Category Theory that are present throughout this write up and the field itself. Firstly, duality in category roughly refers to the process of swapping some aspect of the object, however, statements and results are still true under this involution [5]. This usually arises from swapping domains and codomains of arrows (fundamentally, we are just working in the opposite categories instead, this will be defined later on). This write up will only touch on basic abstract dualities, such as those in universal properties and functors.

Additionally, commutative diagrams are mainly helpful visual tools to map out the structures that we come across, and to communicate and encapsulate fundamental properties that define categorical concepts or universal properties. To say a diagram commutes means the composition of arrows through different paths from one end point to another are equal to each other.

1.2 Categories

Definition 1.2.1 (Category)

A Category, $\mathcal{C}$, consists of [11, p. 3]:

- *a collection of objects $\text{Obj}(\mathcal{C}) = X, Y, Z, \dots$*
- *and a collection of arrows (also called morphisms) $\text{Arr}_{\mathcal{C}}(X, Y)$ for all objects X, Y in $\mathcal{C}$ (When convenient, the collection of all arrows will be denoted $\text{Arr}(\mathcal{C})$), where*
 - *each $f \in \text{Arr}(\mathcal{C})$ has a domain and codomain*
 - *each object $X \in \text{Obj}(\mathcal{C})$ has an identity arrow $\mathbb{1}_X : X \rightarrow X$*

– composition on the arrows is defined as,

$$\begin{aligned}\circ : \text{Arr}_{\mathcal{C}}(X, Y) \times \text{Arr}_{\mathcal{C}}(Y, Z) &\rightarrow \text{Arr}_{\mathcal{C}}(X, Z) \\ (f, g) &\mapsto g \circ f\end{aligned}$$

that is, for any two arrows f, g in $\text{Arr}(\mathcal{C})$ such that the domain of f and codomain of g match, their composite $g \circ f$ is also contained in $\text{Arr}(\mathcal{C})$

such the following two axioms hold:

1. *Left and Right Identities:*

for $f : X \rightarrow Y$, $f \circ \mathbb{1}_X = f$ and $\mathbb{1}_Y \circ f = f$

2. *Associativity of Composition:*

$$(h \circ g) \circ f = h \circ (g \circ f)$$

Remark 1.2.2

In particular, when we define categories, the arrows between these objects tend to be maps that preserve the structure of the objects in the category. This is evident in the examples below.

Example 1.2.3 (Examples of Categories)

1. Category of Sets, **Set**.

- $\text{Obj}(\mathbf{Set})$: Sets
- $\text{Arr}(\mathbf{Set})$: Functions (can be any function between sets)

2. Category of Vector Spaces over some field K , **Vect_K**.

- $\text{Obj}(\mathbf{Vect}_K)$: Vector Spaces over K
- $\text{Arr}(\mathbf{Vect}_K)$: Linear maps

3. Category of Groups, **Grp**

- $\text{Obj}(\mathbf{Grp})$: Groups
- $\text{Arr}(\mathbf{Grp})$: Group homomorphisms

4. Category of Conformally Equivalent Spaces

- Obj : Open Sets of $\mathbb{C}^n$ for fixed n [1]
- Arr : Biholomorphic maps (also known as conformal maps)

With some work and background knowledge, it is evident that all of these examples satisfy the two axioms. We can also restrict the maps and objects we consider as needed, as long as they satisfy the axioms. For example, we can narrow down **Grp** to only consider Abelian Groups, **Ab**, where the arrows will also be restricted to group homomorphisms between Abelian Groups. Thus, **Ab** is a subcategory of **Grp**.

Certain mathematical objects can be formalized as Categorical constructions themselves, where we mimic the algebraic structure and properties of these objects within the structure of an abstract category. This is the process of “Horizontal Categorification” [9].

Recall that a monoid consists of a set of elements M equipped with a binary operation $\cdot : M \times M \rightarrow M$, such that the identity w.r.t. $\cdot$, $e \in M$, and $\cdot$ is associative. Groups additionally require the existence of inverses, for all $m \in M$, there exists $m^{-1} \in M$ such that $m \cdot m^{-1} = m^{-1} \cdot m = e$.

Example 1.2.4 (Monoids and Groups as Categories)

Consider a one-object category, X , where the object is the monoid M , and each arrow is an element of M , $\text{Arr}(X) = M$. The identity arrow is simply $e \in M$, where $e \cdot m = m \cdot e = m$. The composition of arrows $m, n \in \text{Arr}(X)$ is just the element $m \cdot n \in \text{Arr}(X)$, thus satisfying category axioms.

As for groups, since each element has an inverse, all morphisms are isomorphisms.

We can generalize the latter concept from one-object categories to general categories, that is, a **Groupoid** is a category such that all morphisms are isomorphisms.

Example 1.2.5 (The Natural Numbers $\mathbb{N}$ as Categories?)

This is an interesting case, since we are dealing with two binary operations, one forming a group, and the other forming a monoid. Therefore, we actually can't define it using the usual construction of a category. This will arise again later on in the natural transformation section.

Definition 1.2.6 (Opposite Category)

The Opposite Category, denoted $\mathcal{C}^{op}$, is the “dual” of a Category $\mathcal{C}$, where $\text{Obj}(\mathcal{C}^{op}) = \text{Obj}(\mathcal{C})$ and the domain and codomain of arrows are flipped, ie. $f : X \rightarrow Y \in \text{Arr}(\mathcal{C})$ is associated with $f : Y \rightarrow X \in \text{Arr}(\mathcal{C}^{op})$

Note that this still satisfies Category axioms as in Definition 1.2.1. This is actually a “natural” construction, in the sense that given any category, we can automatically construct its dual Opposite category simply by reversing the arrows. Note that for $\mathcal{C}$ a groupoid, $\mathcal{C} = \mathcal{C}^{op}$.

Definition 1.2.7 (Product Category)

The Product Category $\mathcal{C} \times \mathcal{D}$ between two categories $\mathcal{C}$, $\mathcal{D}$ is a category where $\text{Obj}(\mathcal{C} \times \mathcal{D}) = \{(X, Y) | X \in \text{Obj}(\mathcal{C}), Y \in \text{Obj}(\mathcal{D})\}$ and $\text{Arr}(\mathcal{C} \times \mathcal{D}) = \{(f_C, f_D) | f_C \in \text{Arr}(\mathcal{C}), f_D \in \text{Arr}(\mathcal{D})\}$ [10]

1.3 Functors

Definition 1.3.1 (Functor)

A Functor F is an arrow between two categories $\mathcal{C}, \mathcal{D}$, whose data consists of the images [11, p. 13]:

- $F(X) \in \text{Obj}(\mathcal{D})$ of each object $X \in \text{Obj}(\mathcal{C})$ under F
- $F(f) : F(X) \rightarrow F(Y) \in \text{Arr}(\mathcal{D})$ of each arrow $f : X \rightarrow Y \in \text{Arr}(\mathcal{C})$ under F

such that it satisfies two axioms:

1. Identity maps to identity:
for all $X \in \text{Obj}(\mathcal{C})$ $F(1_X) = 1_{F(X)}$
2. Commutes with composition:
 $F(g \circ f) = F(g) \circ F(f)$

Remark 1.3.2

In the same way the arrows of a category are structure preserving maps, functors also preserve categorical structure.

In particular, the functor axioms imply that if we were to treat functors as arrows in a category, it turns out it satisfies the three arrow conditions and two axioms listed in Definition 1.2.1 above. As a result, we obtain **Cat**, the Category of (small) categories, where its objects are small categories and arrows are the functors between them.

Remark 1.3.3 (Small categories)

Here, “small” means the collection of all arrows is on the level of set, which consequently imply the collection of objects is also a set, as opposed to a “class”. This avoids running into issues similar to defining the set of all sets (paradox!). In our later discussion of Hom-functors in Remark 1.3.8, we loosen the requirement to “locally small” categories, where Hom-sets between any pair of objects are sets [11, p. 7].

In the previous section, we discussed opposite categories, emphasizing that a category in which all its arrows are reversed will still satisfy category axioms. This leads to two main types of functors.

Definition 1.3.4 (Contravariant Functor)

A Contravariant Functor is a functor which reverses the directions of all the arrows in its domain category, that is, it's a functor from $\mathcal{C}^{\text{op}}$ to any target category $\mathcal{D}$ [11, p. 17]. For example, given $f : X \rightarrow Y \in \text{Arr}(\mathcal{C})$, $F(f)$ is a map $F(f) : F(Y) \rightarrow F(X)$, where F is the contravariant functor.

Definition 1.3.5 (Covariant Functor)

In contrast to contravariant functors, the Covariant Functor is just the typical definition of a functor that does not reverse the direction of the arrows.

If $\mathcal{C}$ is a groupoid, then contravariant functors can be turned into and covariant functors and vice versa, due to all arrows being invertible.

Example 1.3.6 (Forgetful functors)

Forgetful functors generally refer to functors where objects lose structure under it. For instance, the functor $\mathbf{Grp} \rightarrow \mathbf{Set}$ only recovers the underlying sets of groups, and arrows in $\mathbf{Grp}$ are simply mapped to functions defined on the underlying set [11, p. 13].

Example 1.3.7 (Hom-functors)

Another example are Hom-functors, they technically are **Bifunctors** whose domain is a product category (Definition 1.2.7), $\mathrm{Hom}(-, -) : \mathcal{C}^{op} \times \mathcal{C} \rightarrow \mathbf{Set}$ [8]. It maps,

- *Obj*:

$$(X, X') \rightarrow \mathrm{Hom}_{\mathcal{C}}(X, X')$$

where $\mathrm{Hom}_{\mathcal{C}}(X, X')$ are Hom-Sets, ie. the set of arrows between X, X' in $\mathcal{C}$

- *Arr*:

$$\begin{aligned} [(f^{op}, g) : (X, X') \rightarrow (Y, Y')] &\rightarrow [\mathrm{Hom}_{\mathcal{C}}(X, X') \rightarrow \mathrm{Hom}_{\mathcal{C}}(Y, Y')] \\ (h : X \rightarrow X') &\rightarrow g \circ h \circ f : Y \rightarrow X \rightarrow X' \rightarrow Y' \end{aligned}$$

Note that defining the first component of the domain as the opposite category is technically important for the composition $g \circ h \circ f$, since we can't directly use $f : Y \rightarrow X$ when we consider arrows $(X, X') \rightarrow (Y, Y')$.

Remark 1.3.8

Hom-functors will be of particular importance as we discuss the Yoneda Lemma in Theorem 2.1.1.

In particular, we will be fixing one of the components,

$$\mathrm{Hom}(X, -) : \mathcal{C} \rightarrow \mathbf{Set}$$

$$\mathrm{Hom}(-, X) : \mathcal{C}^{op} \rightarrow \mathbf{Set}$$

which represents the set of all arrows mapping out of X or into X . By definition, the former is a covariant functor of $\mathcal{C}$, taking morphisms $f : Y \rightarrow Z \in \mathrm{Arr}(\mathcal{C})$ to the **pushforward**

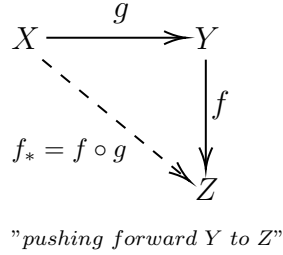
$$\begin{aligned} f_* : \mathrm{Hom}(X, Y) &\rightarrow \mathrm{Hom}(X, Z) \\ g &\mapsto f \circ g \end{aligned}$$

which takes f and postpends it with any map $g : X \rightarrow Y$, that is $f_*(g) = f \circ g$. The latter is a contravariant functor of $\mathcal{C}$, taking morphisms $f^{op} : Z \rightarrow Y \in \text{Arr}(\mathcal{C}^{op})$ to the **pullback**

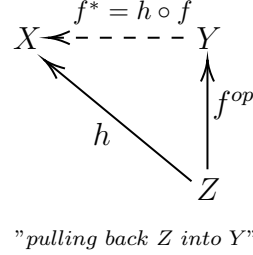
$$\begin{aligned} f^* : \text{Hom}(Z, X) &\rightarrow \text{Hom}(Y, X) \\ h &\mapsto h \circ f \end{aligned}$$

which takes f and prepends it with any map $h : Z \rightarrow X$, that is $f^*(h) = h \circ f$.

Covariant :



Contravariant :



Example 1.3.9 (Dual, Double Dual Vector Spaces)

Furthermore, Dual Vector Spaces can be thought of as contravariant and covariant constructions.

In particular, recall the definition of dual vector spaces:

Definition 1.3.10 (Dual space)

Given a Vector Space V over F , the dual space V^* is the set of all linear maps $f : V \rightarrow F$. Using the Hom notion above, $V^* = \text{Hom}(V, F)$.

For instance, the projection map taking $v = (v_1, \dots, v_n) \in V$ to v_1 is an element of V^* . We further define the double dual space V^{**} , ie. the dual space of V^* .

Definition 1.3.11 (Double dual space)

Given a Vector Space V over F , the double dual space V^{**} is the set of all linear maps $\beta : V^* \rightarrow F$, ie $V^{**} = \text{Hom}(V^*, F)$.

This follows naturally by applying Def 1.3.10 to V^* . One way to think about V^{**} is, its elements are "evaluation maps" of $\beta \in V^{**}$. For instance, if we fix some $v_0 \in V$, this induces a map $\beta \rightarrow \beta(v_0) \in F$.

Let $-^*$ and $-^{**}$ be the dual and double dual operations respectively, ie, defined as maps that take vector spaces to their duals and double duals,

$$\begin{aligned} -^* : \mathbf{Vect}_F &\rightarrow \mathbf{Vect}_F \\ V &\mapsto V^* \\ -^{**} : \mathbf{Vect}_F &\rightarrow \mathbf{Vect}_F \\ V &\mapsto V^{**} \end{aligned}$$

Consider $\mathbf{Vect}_F$ (over field F) and treat $-^*$ and $-^{**}$ as functors from $\mathbf{Vect}_F$ to itself, we now want to show that they are contravariant and covariant functors respectively. For any linear map $T : V \rightarrow W$, how can we construct a mapping between V^* and W^* ? Well, given that $T : V \rightarrow W$, we can prepend any $g \in W^*$ as follows: $g \circ T : V \rightarrow W \rightarrow R$, therefore $g \circ T \in V^*$. As a result,

$$\begin{aligned} T^* : W^* &\rightarrow V^* \\ g &\mapsto g \circ T \end{aligned}$$

is a contravariant functor! Applying the same construction to get the double dual, we have that,

$$\begin{aligned} T^{**} = (T^*)^* : (V^*)^* &\rightarrow (W^*)^* \\ \alpha &\mapsto \alpha \circ T^* \end{aligned}$$

where given any $\alpha \in V^{**}$, we prepend $T^* : W^* \rightarrow V^*$. Drawing back on the intuition above, α is the evaluation map on V^* , which plugs some fixed $v_0 \in V$ into a map $f \in V^*$, ie. $\alpha(-) = -(v_0)$. Then $\alpha \circ T$ is a map $\alpha(- \circ T) = (- \circ T)(v_0)$, effectively "pulling back" some $g \in W^*$ to a map in V^* so that it can be evaluated on some element of V . As a result, T^{**} is a covariant functor!

This example will come in handy in the next section. Lastly, the following two types of functors are also prevalent, and are analogs of surjective and injective functions for functors [11, p. 30]

Definition 1.3.12 (Full functor)

A functor $F : \mathbf{C} \rightarrow \mathbf{D}$ is full if the mapping on arrows is surjective, that is,

$$F(\text{Hom}(X, Y)) = \text{Hom}(F(X), F(Y))$$

for all $X, Y \in \text{Obj}(\mathbf{C})$

Definition 1.3.13 (Faithful functor)

A functor $F : \mathbf{C} \rightarrow \mathbf{D}$ is faithful if the mapping on arrows is injective, that is

$$F(\text{Hom}(X, Y)) \hookrightarrow \text{Hom}(F(X), F(Y))$$

for all $X, Y \in \text{Obj}(\mathbf{C})$

1.4 Natural Transformations

A natural transformation is an arrow between functors that preserve certain categorical structures.

Definition 1.4.1 (Natural Transformation)

Given categories $\mathcal{C}, \mathcal{D}$, and functors F, G , the data of a natural transformation $\Phi : F \rightarrow G$ consists of [11, p. 23]:

- the data of the functors (see Definition 1.3.1)
- the components of the natural transformation is defined as the collection of $\Phi_X : F(X) \rightarrow G(X) \in \mathcal{D}$ for each object $X \in \mathcal{C}$

such that the following diagram commutes:

$$\begin{array}{ccc} F(X) & \xrightarrow{\Phi_X} & G(X) \\ F(f) \downarrow & \circ & \downarrow G(f) \\ F(Y) & \xrightarrow{\Phi_Y} & G(Y) \end{array}$$

Example 1.4.2 (Double dual vector spaces)

An example of a natural transform lies in double dual vector spaces.

Recall the construction of taking dual and double dual of vector spaces as functors in 1.3.10 and 1.3.11 above. Now, we want to construct a map $V \rightarrow V^{**}$ s.t. it is a natural transformation. We can define it as follows:

$$\begin{aligned} \eta : V &\rightarrow V^{**} \\ v &\mapsto (f \mapsto f(v)) \\ \text{where } \eta(v) : V^* &\rightarrow F \\ f &\mapsto f(v) \end{aligned}$$

Thus, $(\eta(v)(f)) \in F$. Note that this is not the only way to define a map from V to V^{**} . Now, we show that η is indeed a natural transformation from the identity functor to the double dual functor. Given $V, W \in \text{Obj}(\mathbf{Vect}_F)$ and $T : V \rightarrow W \in \text{Arr}(\mathbf{Vect}_F)$, consider the two functors, $\mathbb{1}, -^{**} : \mathbf{Vect}_F \rightarrow \mathbf{Vect}_F$. Let $\Phi_V = \eta_V : \mathbb{1} \rightarrow -^{**}$. Applying the functors $\mathbb{1}$ and $-^{**}$ to the diagram in Def 1.4.1, we have,

$$\begin{array}{ccc} \mathbb{1}(V) & \xrightarrow{\Phi_V} & (V)^{**} \\ \mathbb{1}(T) \downarrow & \circ & \downarrow (T)^{**} \\ \mathbb{1}(W) & \xrightarrow{\Phi_W} & (W)^{**} \end{array} \quad \rightsquigarrow \quad \begin{array}{ccc} V & \xrightarrow{\eta_V} & V^{**} \\ T \downarrow & \circ & \downarrow T^{**} \\ W & \xrightarrow{\eta_W} & W^{**} \end{array}$$

If this diagram commutes, then η satisfies the definition of a natural transform. We first compute what each of the four maps do: (note that f, g, α are arbitrary elements in V^*, W^*, V^{**})

$$\begin{aligned}\eta_V(v) &= (f \mapsto f(v)) \in V^{**} \\ \eta_W(w) &= (g \mapsto g(w)) \in W^{**} \\ T(v) &= Tv \in W \\ T^{**}(\alpha) &= \alpha(g' \circ T) \in W^{**}\end{aligned}$$

Then we can check commutativity, assume we fix some $g \in W^*$. Along η_V then T^{**} :

$$\begin{aligned}T^{**}(\eta_V(v)) &= (\eta_V(v))(g \circ T) \\ &= (f \mapsto f(v)), f = g \circ T \\ &= (g \circ T)(v) \\ &= gTv\end{aligned}$$

Along T then η_W :

$$\begin{aligned}\eta_W(Tv) &= (g \mapsto g(Tv)) \\ &= gTv\end{aligned}$$

Therefore, we have that the diagram commutes, and η is a natural transformation!

Remark 1.4.3 (N-morphisms and Functor Categories)

*To go off on a tangent, in **Cat**, a natural transformation is considered called a 2-morphism, while functors are 1-morphisms (and arrows are 0-morphisms). This naming convention is indicative of the higher dimensions that mathematicians keep building, by considering arrows between arrows between arrows between ... etc. One way we can formalize these constructions is with the functor category.*

Definition 1.4.4 (Functor Category)

The Functor Category, $\mathcal{D}^{\mathcal{C}}$, consists of $\text{Obj}(\mathcal{D}^{\mathcal{C}})$, the collection of all functors $F : \mathcal{C} \rightarrow \mathcal{D}$ between categories $\mathcal{C}, \mathcal{D}$, and $\text{Arr}(\mathcal{D}^{\mathcal{C}})$, the collection of the natural transformations between these functors [7].

In particular, we can keep constructing higher dimensional categories too. The notion of a 2-category arises when we expand the notion of categories to include arrows between arrows. For instance, a 2-category **Cat** is a Category where we “enrich” a category over **Cat**. Formalizing rings, such as **N** above, as categories actually lead to the enriched Category over **Ab**, the Category of Abelian Groups. We can keep adding more dimensions, leading to n-Categories, each of which are much richer and more complex in structure than the prior.

1.5 Universal Properties

Universal properties are a fundamental concept in category theory. The point of category theory is to apply the same frameworks or patterns in different areas of mathematics, these frameworks are precisely what universal properties are. When we derive results in these general frameworks, by applying them to specific mathematical constructs, we automatically get analogous results for these constructs. Therefore, constructions and proofs may be translated across fields, offering new perspectives on existing problems. Additionally, universal properties characterize objects up to unique isomorphism, and its consequences we will see below.

Two objects defined by their universal properties that will come up in our discussion of the Yoneda Lemma are given as follows [12],

Definition 1.5.1 (Product)

Given a family $(X_i)_{i \in I}$ in a category $\mathcal{C}$, the Product is $X = \prod X_i$ (also denoted with $\otimes$ or $\times$), equipped with maps $\pi_i : X \rightarrow X_i$ for all $i \in I$, s.t. for all Y and maps $f_i : Y \rightarrow X_i$, there exists a unique $f : Y \rightarrow X$, s.t. the following diagram commutes for all i :

$$\begin{array}{ccc} X & \xrightarrow{\pi_j} & X_j \\ \uparrow \exists! f & \circlearrowleft & \nearrow f_j \\ Y & & \end{array}$$

Definition 1.5.2 (Coproduct)

Given a family $(X_j)_{j \in J}$ in a category $\mathcal{C}$, the Coproduct is $X = \coprod X_j$ (also denoted with $\oplus$), equipped with maps $i_j : X_j \rightarrow X$ for all $j \in J$, s.t. for all Y and maps $g_j : X_j \rightarrow Y$, there exists a unique $g : X \rightarrow Y$, s.t. the following diagram commutes for all j :

$$\begin{array}{ccc} X & \xleftarrow{i_j} & X_j \\ \downarrow \exists! g & \circlearrowleft & \searrow g_j \\ Y & & \end{array}$$

Since both of these are universal constructions, for any X' that is also a product or coproduct, there exists an isomorphism $h : X \rightarrow X'$ s.t. $\pi'_i = \pi \circ h$ and $i'_j = h \circ i_j$ respectively. This is what it means to be unique up to isomorphism.

Remark 1.5.3

It is worth noting that a product or coproduct does not always exist in every category! Same goes for other universal properties, certain categories will have certain universal properties depending on the properties of the objects it contains.

Example 1.5.4 (Cartesian Product)

The cartesian product is an example of the product in **Set**, whose elements are ordered tuples consisting of elements of each set in the product. π_i is the projection map $\pi_i(x) = x_i$, mapping a tuple x to the element of its i -th index.

2 The Yoneda Lemma

The Yoneda Lemma is one of the most fundamental results in Category Theory. On an high level overview, it says that everything we need to know about an object is encoded in its relationships with other objects.

2.1 Statement

The essence of the Yoneda Lemma lies in the following correspondence:

Theorem 2.1.1 (Yoneda Lemma [3])

The Yoneda Lemma states that for any X, Y in a Category $\mathcal{C}$, there is a natural isomorphism between:

- the set of arrows between X, Y , $\text{Hom}(X, Y)$
- and the set of natural transformations $\text{Hom}(\text{Hom}(-, X), \text{Hom}(-, Y))$ between the hom-functor of X and the hom-functor of Y

$$\text{Hom}(X, Y) \cong \text{Hom}(\text{Hom}(-, X), \text{Hom}(-, Y))$$

Remark 2.1.2

Note that the above is the contravariant construction (Definition 1.3.5), we can similarly formalize the Yoneda Lemma with covariant functors

$$\text{Hom}(X, Y) \cong \text{Hom}(\text{Hom}(Y, -), \text{Hom}(X, -))$$

Let's breakdown this construction. We begin by plugging in any object $Z \in \mathcal{C}$ into the isomorphism above, which gives us the correspondence $\text{Hom}(X, Y) \cong \text{Hom}(\text{Hom}(Z, X), \text{Hom}(Z, Y))$. The motivation here lies in post or pre-composition of maps. Given a map $f : X \rightarrow Y$, by postcomposing maps $h : Z \rightarrow X \in \text{Hom}(Z, X)$ by f , we obtain maps $g = f \circ h : Z \rightarrow Y \in \text{Hom}(Z, Y)$.

How do we make sense of this correspondence? That is, given a natural transformation $\mathcal{F}_- : \text{Hom}(-, X) \rightarrow \text{Hom}(-, Y)$, how do we obtain a map $f : X \rightarrow Y$. Conversely, given a map $f : X \rightarrow Y$, how do we obtain a natural transformation $\mathcal{F}_- : \text{Hom}(-, X) \rightarrow \text{Hom}(-, Y)$ that applies to any object X ?

The former is fairly easy to obtain. We know we have $\mathcal{F}_-$, then plugging in X gives us $\mathcal{F}_X : \text{Hom}(X, X) \rightarrow \text{Hom}(X, Y)$. Therefore, applying the natural transform on the identity homomorphism, $\mathcal{F}_X(1_X) = f \in \text{Hom}(X, Y)$ gives us precisely the morphism $f : X \rightarrow Y$, which is unique by the Yoneda Lemma.

We rely on the motivation above to argue for the converse. Given a map $f : X \rightarrow Y$, we can define a natural transformation $\Phi : \text{Hom}(-, X) \rightarrow \text{Hom}(-, Y)$ where $\Phi(g) = f \circ g$ where g is an element of the hom-functor $\text{Hom}(-, X)$, and the codomains of g and $\Phi(g)$ match. The fundamental idea here is exactly taking the pushforward of maps in $\text{Hom}(-, X)$ for arbitrary elements $-$. Φ is again unique by the Yoneda Lemma.

The same argument applies for the covariant version. For the former, we plug in Y to obtain: $\mathcal{F}_Y : \text{Hom}(Y, Y) \rightarrow \text{Hom}(X, Y)$, thus $\mathcal{F}_Y(1_Y) : X \rightarrow Y$. For the latter, we instead obtain the natural transformation $\Phi : \text{Hom}(Y, -) \rightarrow \text{Hom}(X, -)$ by precomposing $\Phi(g) = g \circ f$ for $g \in \text{Hom}(Y, -)$, thus the domains of g and $\Phi(g)$ match. By the Yoneda Lemma, both of these constructions are unique.

Remark 2.1.3 (General version of the Yoneda Lemma)

In fact, not only does this hold for $\text{Hom}(-, Y)$, but the Yoneda Lemma also holds for any contravariant functor $G : \mathcal{C}^{op} \rightarrow \mathbf{Set}$ and thus generalizes to the following correspondence for any object X in $\mathcal{C}$:

$$G(X) \cong \text{Hom}(\text{Hom}(-, X), G(-))$$

and in the covariant case, for any covariant functor $F : \mathcal{C} \rightarrow \mathbf{Set}$,

$$F(X) \cong \text{Hom}(\text{Hom}(X, -), F(-))$$

2.2 Consequences

Lemma 2.2.1 (Yoneda Embedding)

Recall the discussion on functor categories in Definition 1.4.4. A nice consequence of the Yoneda Lemma is in regards to the functor category whose objects are Hom-functors. From Example 1.3.7, we know that the hom-functors $\text{Hom}(-, X)$ as objects live in the functor category $\mathbf{Set}^{\mathcal{C}^{op}}$. Naturally, the arrows in this category are mappings $\text{Hom}(-, X) \rightarrow \text{Hom}(-, Y)$, which are precisely natural transformations between these Hom-sets in the standard treatment. Therefore, this induces a functor,

$$\begin{aligned} F : \mathbf{C} &\rightarrow \mathbf{Set}^{\mathcal{C}^{op}} \\ X &\rightarrow \text{Hom}(-, X) \\ (f : X \rightarrow Y) &\rightarrow (f_* : \text{Hom}(-, X) \rightarrow \text{Hom}(-, Y)) \end{aligned}$$

The notation for f_* mimicking the pullback is once again deliberate, since we are simply post-composing by f . We can check that this truly satisfies the definition of a functor by

checking the diagram in Definition 1.3.1 commutes. Most importantly, the Yoneda Lemma ensures that each morphism f will uniquely induce the natural transformation f_* , exactly as we described above.

Definition 2.2.2 (Presheaves and Category of Presheaves)

*The functor category, $\mathbf{Set}^{C^{op}}$, is called the **Category of presheaves** (of (C)), and its elements are called **presheaves**. Note that objects of the form $\mathrm{Hom}(-, X)$ are only a subcollection of all of its elements, presheaves of this form are called **representable functors**.*

A similar construction follows for the covariant Hom-functor $\mathrm{Hom}(X, -)$ and the functor category $\mathbf{Set}^C$. In this case, elements $\mathrm{Hom}(X, -)$ are **copresheaves**.

Recall the notions of full and faithful in Definitions 1.3.12 and 1.3.13. Since the Yoneda Lemma gives us an isomorphism between $\mathrm{Hom}(X, Y)$ and $\mathrm{Hom}(\mathrm{Hom}(-, Y), \mathrm{Hom}(-, X))$, we have that the functor F is full and faithful on its image, which is a subcategory of $\mathbf{Set}^{C^{op}}$. Thus, F effectively embeds the category $\mathbf{C}$ into its category of presheaves $\mathbf{Set}^{C^{op}}$. This is the Yoneda Embedding [2]. It is precisely in this embedding that we can think of an object X as being represented by $\mathrm{Hom}(-, X)$, because we have this full and faithful embedding into the subcategory of presheaves of these representable functors.

Lemma 2.2.3 (Objects are isomorphic if their hom-sets are (naturally) isomorphic)

If $\mathrm{Hom}(-, A) \cong \mathrm{Hom}(-, B)$, then $A \cong B$

This is a direct consequence of the Yoneda Lemma, since there is a unique map $A \rightarrow B$ corresponding to the natural transform $\mathrm{Hom}(-, A) \cong \mathrm{Hom}(-, B)$. Since the latter is invertible, we must also have that the former has a unique inverse.

2.3 Examples

Example 2.3.1 (Associativity and Commutativity of the Product)

Assuming we have the product of sets is associative and commutative, proving that the product and coproduct constructions (Definitions 1.5.1, 1.5.2) are associative and commutative (up to isomorphism) is a simple application of the Yoneda Lemma.

However, the lemma operates on the level of hom-sets, so how can we argue about the properties of the product and coproduct as its own object?

By the universal property of product, for any object X , there is a unique map from X to $R \otimes S$ satisfying the commutative diagram. This gives us an isomorphism between maps into $R \otimes S$ and the product of maps into R and S . Also, by the commutativity of product of sets,

we have the following:

$$\begin{aligned}
\text{Hom}(-, R \otimes S) &\cong \text{Hom}(-, R) \times \text{Hom}(-, S) \\
&\cong \text{Hom}(-, S) \times \text{Hom}(-, R) \\
&\cong \text{Hom}(-, S \otimes R)
\end{aligned}$$

In particular, we have an isomorphism, $\text{Hom}(-, R \otimes S) \cong \text{Hom}(-, S \otimes R)$ on the level of hom-sets, that is, we have a natural isomorphism $\text{Nat}(\text{Hom}(-, R \otimes S) \cong \text{Hom}(-, S \otimes R))$. Finally, by the Yoneda Lemma, this induces a map in $\text{Hom}(R \otimes S, S \otimes R)$, specifically an isomorphism. Therefore, $R \otimes S \cong S \otimes R$.

For associativity, a similar process applies using the universal property and associativity of the set product.

$$\begin{aligned}
\text{Hom}(-, (R \otimes S) \otimes T) &\cong \text{Hom}(-, R \otimes S) \times \text{Hom}(-, T) \\
&\cong (\text{Hom}(-, R) \times \text{Hom}(-, S)) \times \text{Hom}(-, T) \\
&\cong \text{Hom}(-, R) \times (\text{Hom}(-, S) \times \text{Hom}(-, T)) \\
&\cong \text{Hom}(-, R) \times \text{Hom}(-, S \otimes T) \\
&\cong \text{Hom}(-, R \otimes (S \otimes T))
\end{aligned}$$

Thus by the Yoneda Lemma, $(R \otimes S) \otimes T \cong R \otimes (S \otimes T)$.

As for the coproduct, we use the contravariant version of the Yoneda Lemma and apply the same principles,

$$\begin{aligned}
\text{Hom}(R \oplus S, -) &\cong \text{Hom}(S, -) \times \text{Hom}(R, -) \\
&\cong \text{Hom}(R, -) \times \text{Hom}(S, -) \\
&\cong \text{Hom}(S \oplus R, -)
\end{aligned}$$

and likewise,

$$\begin{aligned}
\text{Hom}((R \oplus S) \oplus T, -) &\cong \text{Hom}(T, -) \times \text{Hom}(R \oplus S, -) \\
&\cong \text{Hom}(T, -) \times (\text{Hom}(S, -) \times \text{Hom}(R, -)) \\
&\cong (\text{Hom}(T, -) \times \text{Hom}(S, -)) \times \text{Hom}(R, -) \\
&\cong \text{Hom}(S \oplus T) \times \text{Hom}(R, -) \\
&\cong \text{Hom}(R \oplus (S \oplus T), -)
\end{aligned}$$

By the Yoneda Lemma, $R \oplus S \cong S \oplus R$ and $(R \oplus S) \oplus T \cong R \oplus (S \oplus T)$

Example 2.3.2 (Cayley's Theorem)

The Yoneda Lemma technically stems from generalizing Cayley's Theorem, which states that

every group G is isomorphic to a subgroup of the Symmetric group $Sym(G)$, the group of bijections from G to G [11, p. 61].

Firstly, we treat G as the single element in the one-object category as defined in Example 1.2.4. By the requirements for the Yoneda Lemma, the contravariant functor F is in the functor category $\mathbf{Set}^{G^{op}}$ [6]. Objects in this category are G -sets, defined as a set X equipped with the operation of the left action of G on X ,

$$\begin{aligned} \cdot : G \times X &\rightarrow X \\ (g, x) &\mapsto g \cdot x \end{aligned}$$

Consequently, the arrows of this category are G -equivalent maps, that preserve the $\cdot$ operation, that is, for $f : X \rightarrow Y \in Arr(\mathbf{Set}^{G^{op}})$, $\forall g \in G, \forall x \in X, f(g \cdot x) = g \cdot f(x)$.

By setting this functor to be the contravariant Hom-functor, $F = Hom(-, G)$, and applying the Yoneda Lemma, we retrieve

$$Hom(G, G) \cong Hom(Hom(-, G), Hom(-, G))$$

where $-$ is any object in the category.

Note that $Hom(G, G)$ simply consists of the elements of G in this one-object category, thus $Hom(G, G) = G$. Additionally, by our previous discussion, $Hom(-, G)$ are G -sets in the functor category.

Yoneda gives us an isomorphism that takes each element $g \in G$ to a natural transformation, ie. G -equivariant maps $\eta_g : Hom(-, G) \rightarrow Hom(-, G)$. When evaluated on the only object in the category, this map turns out to be

$$\begin{aligned} \eta_g : Hom(-, G) &\rightarrow Hom(-, G) \\ h &\mapsto g \cdot h \end{aligned}$$

which is an automorphism of G [4]. Thus $\{\eta_g | g \in G\} = Aut(G) \subseteq Sym(G)$. Finally, $G \cong Aut(G) \subseteq Sym(G)$.

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References

- [1] *Biholomorphism*. Sept. 2023. URL: en.wikipedia.org/wiki/Biholomorphism.
- [2] Tai-Danae Bradley. *The Yoneda Embedding*. Sept. 2017. URL: www.math3ma.com/blog/the-yoneda-embedding.
- [3] Tai-Danae Bradley. *The Yoneda Lemma*. Sept. 2017. URL: www.math3ma.com/blog/the-yoneda-lemma.
- [4] Travis Dirlle. *Category Theory: The Yoneda Lemma and Cayley's Theorem*. June 2017. URL: www.youtube.com/watch?v=-OB1t1iZZpo.
- [5] nLab. *duality*. Nov. 2024. URL: ncatlab.org/nlab/show/duality.
- [6] nLab. *equivariant*. Sept. 2022. URL: ncatlab.org/nlab/show/equivariant.
- [7] nLab. *functor category*. May 2023. URL: ncatlab.org/nlab/show/functor-category.
- [8] nLab. *hom-functor*. June 2022. URL: ncatlab.org/nlab/show/hom-functor.
- [9] nLab. *horizontal categorification*. Jan. 2015. URL: ncatlab.org/nlab/show/horizontal+categorification.
- [10] nLab. *product category*. June 2023. URL: ncatlab.org/nlab/show/product+category.
- [11] Emily Riehl. *Category Theory in Context*. Dover Publications, 2015. ISBN: 9780486809038.
- [12] Joseph Sullivan. *Products and Coproducts*. Dec. 2021. URL: partiallyordered.com/posts/products-and-coproducts.